Mrinal Lobo

DevOps Engineer
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Summary

DevOps Engineer with 4 years of experience building and running production Kubernetes platforms on Azure and AWS, with Terraform, GitOps, CI/CD, observability, security and incident response. AZ-104 and GH-200 certified.

Technical Skills

Cloud: Azure (AKS, VMs, SQL Server, Key Vault), AWS (EKS, ECS, EC2, EventBridge, Bedrock), GCP (GKE)
Kubernetes & GitOps: Kubernetes, Docker, Helm, Flux, Velero, Kubernetes operators
IaC & CI/CD: Terraform, Ansible, GitHub Actions, Azure DevOps, Jenkins, HashiCorp Vault, Git
Networking & Security: Load balancers, WAF, Route 53 (DNS), federated credentials, image hardening, GDPR
Observability: Prometheus, Grafana, OpenTelemetry, Loki, Alertmanager, Elastic Stack (ELK)
Languages & Data: Python, Bash, SQL, Linux, Kafka
ML & GenAI: scikit-learn, MLflow, LLMs, RAG, vLLM, Ollama, Amazon Bedrock

Experience

EG A/S Aug 2023 – Oct 2026
DevOps Engineer *Mangalore, India*

- Built and operated a multi-tenant, highly available **AKS (Kubernetes)** platform for Estata: 8 nodes, ∼**60 .NET microservices** on Kafka, Velero backups and AWS snapshots, meeting a **99% monthly uptime SLA**.
- Shipped continuously through **GitOps (Flux, Helm)** with rolling updates; built CI/CD with **GitHub Actions** and Azure DevOps, cutting build time from **30 to 5 minutes** by sharding jobs.
- **Security**: hardened chiseled and Bitnami images, federated credentials so GitOps reads **Azure Key Vault** secrets with none in Git, and Go-based Kubernetes operators (Vault unseal), under **GDPR** compliance.
- Provisioned MainManager's Azure environments (VMs, SQL Server) with **Terraform** modules and Vault.
- Built observability with **OpenTelemetry, Prometheus, Grafana and Loki**, alerting to Teams via Alertmanager; tracked **SLOs** (error rate, recovery time) and handled **incident response** and postmortems.
- Designed and ran **MestroBot** in production (Scrapy, Elastic Stack; several GB/day from ∼140 energy companies) with ML models for anomaly detection (Isolation Forest), imputation and consumption forecasting.
- In the AI CoE team, built **LLM / GenAI** RAG chatbots on **Amazon Bedrock** (AWS ECS) and served open models with vLLM; **cost optimisation** cut AWS spend by **$4,000+/month** by right-sizing dev clusters and shutting down idle inference machines with EventBridge.

Aphelion Labs Feb 2023 – Jul 2023
Python Engineer *Indore, India (Remote)*

- Built the backend of Woodle AI (AI website builder on OpenAI GPT APIs) as **Python microservices**.
- Ran its **AWS (ECS, EKS)** infrastructure behind **load balancers and WAF** with Route 53 DNS; was **on call**.
- Built REST APIs for a financial services backend, deployed with Docker and GitHub Actions CI/CD.

Glowtouch Technologies Aug 2022 – Jan 2023
Graduate Engineer Trainee *Mangalore, India*

- Maintained and hardened on-premises **Linux** servers for domain hosting (networking, firewalls, fail2ban).
- Took part in migrating CI/CD pipelines from **Jenkins to GitHub Actions**.
- Wrote **Python and Bash** services and automation for the domain-hosting backend.

Projects

Low-Latency Trading Infrastructure | *Terraform, AWS, Prometheus, Grafana, Loki* | GitHub
- Terraform-provisioned AWS host tuned at kernel and network level for 1–3 ms order fills, with latency dashboards.

AI Market Research Platform | *Python, AI agents, DuckDB, LanceDB, Neo4j* | lumen.tradehook.click
- Self-hosted: Hermes-orchestrated LLM agents (MiniMax) research ∼4,500 stocks over 4.5M+ data points.

GKE SRE Platform | *GCP, Terraform, Ansible, Helm, Vault* | GitHub
- Regional GKE cluster built with Terraform and Ansible: Prometheus, Grafana, Alertmanager, Vault, Rook-Ceph.

Education & Certifications

B.E. Electronics and Communication, AJ Institute of Engineering and Technology 2018 – 2022
Certifications: Microsoft Azure Administrator Associate (AZ-104), GitHub Actions (GH-200)